

Endocrine

“If you skew the endocrine system, you lose the pathways to self.”
—Hilary Mantel

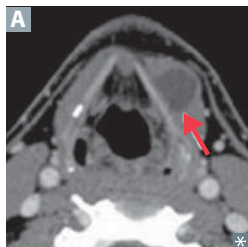
“We have learned that there is an endocrinology of elation and despair, a chemistry of mystical insight, and, in relation to the autonomic nervous system, a meteorology and even . . . an astro-physics of changing moods.”
—Aldous (Leonard) Huxley

“Chocolate causes certain endocrine glands to secrete hormones that affect your feelings and behavior by making you happy.”
—Elaine Sherman, *Book of Divine Indulgences*

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► ENDOCRINE—EMBRYOLOGY

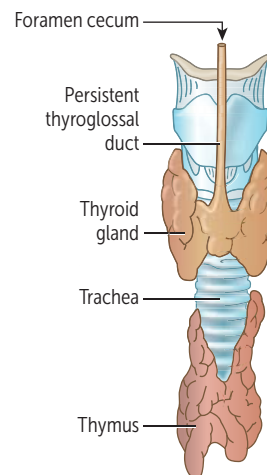
Thyroid development



Thyroid diverticulum arises from floor of primitive pharynx and descends into neck. Connected to tongue by thyroglossal duct, which normally disappears but may persist as cysts or the pyramidal lobe of thyroid. Foramen cecum is normal remnant of thyroglossal duct. Most common ectopic thyroid tissue site is the tongue (lingual thyroid). Removal may result in hypothyroidism if it is the only thyroid tissue present.

Thyroglossal duct cyst **A** presents as an anterior midline neck mass that moves with swallowing or protrusion of the tongue (vs persistent cervical sinus leading to branchial cleft cyst in lateral neck).

Thyroid tissue and parafollicular cells (aka, **C** cells, produce **Calcitonin**) of the thyroid are derived from endoderm.



► ENDOCRINE—ANATOMY

Adrenal cortex and medulla

Adrenal cortex (derived from mesoderm) and medulla (derived from neural crest).

ANATOMY	HISTOLOGY	1° REGULATION BY	HORMONE CLASS	1° HORMONE PRODUCED
	Zona G lomerulosa	Angiotensin II	Mineralocorticoids	Aldosterone
	Zona F asciculata	ACTH, CRH	Glucocorticoids	Cortisol
	Zona R eticularis	ACTH, CRH	Androgens	DHEA
MEDULLA	Chromaffin cells	Preganglionic sympathetic fibers	Catecholamines	Epi, NE

GFR corresponds with **S**alt (mineralocorticoids), **S**ugar (glucocorticoids), and **S**ex (androgens).

“The deeper you go, **the sweeter it gets.**”

Pituitary gland**Anterior pituitary
(adenohypophysis)**

Secretes FSH, LH, ACTH, TSH, prolactin, GH. Melanotropin (MSH) secreted from intermediate lobe of pituitary. Derived from oral ectoderm (Rathke pouch).

- α subunit—hormone subunit common to TSH, LH, FSH, and hCG.
- β subunit—determines hormone specificity.

ACTH and MSH are derivatives of proopiomelanocortin (POMC).

FLAT PiG: FSH, LH, ACTH, TSH, PRL, GH.

B-FLAT: Basophils—FSH, LH, ACTH, TSH.

Acidophils: GH, PRL.

**Posterior pituitary
(neurohypophysis)**

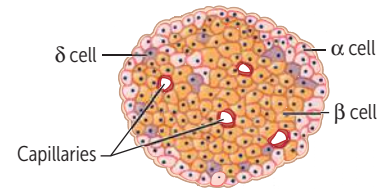
Stores and releases vasopressin (antidiuretic hormone, or ADH) and oxytocin, both made in the hypothalamus (supraoptic and paraventricular nuclei) and transported to posterior pituitary via neurophysins (carrier proteins). Derived from neuroectoderm.

**Endocrine pancreas
cell types**

Islets of Langerhans are collections of α , β , and δ endocrine cells. Islets arise from pancreatic buds.

- α = glucagon (peripheral)
- β = insulin (central)
- δ = somatostatin (interspersed)

Insulin (β cells) inside.

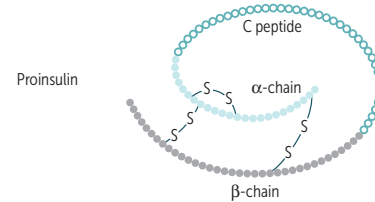


► ENDOCRINE—PHYSIOLOGY

Insulin

SYNTHESIS

Preproinsulin (synthesized in RER) → cleavage of “presignal” → proinsulin (stored in secretory granules) → cleavage of proinsulin → exocytosis of insulin and C-peptide equally. Insulin and C-peptide are ↑ in insulinoma and sulfonylurea use, whereas exogenous insulin lacks C-peptide.



SOURCE

Released from pancreatic β cells.

FUNCTION

Binds **insulin** receptors (tyrosine kinase activity **1**), **inducing** glucose uptake (carrier-mediated transport) **into** insulin-dependent tissue **2** and gene transcription.

Anabolic effects of insulin:

- ↑ glucose transport in skeletal muscle and adipose tissue
- ↑ glycogen synthesis and storage
- ↑ triglyceride synthesis
- ↑ Na⁺ retention (kidneys)
- ↑ protein synthesis (muscles)
- ↑ cellular uptake of K⁺ and amino acids
- ↓ glucagon release
- ↓ lipolysis in adipose tissue

Unlike glucose, insulin does not cross placenta.

Insulin-dependent glucose transporters:

- GLUT4: adipose tissue, striated muscle (exercise can also increase GLUT4 expression)

Insulin-independent transporters:

- GLUT1: RBCs, brain, cornea, placenta
- GLUT2 (bidirectional): β islet cells, liver, kidney, small intestine
- GLUT3: brain, placenta
- GLUT5 (fructose): spermatocytes, GI tract

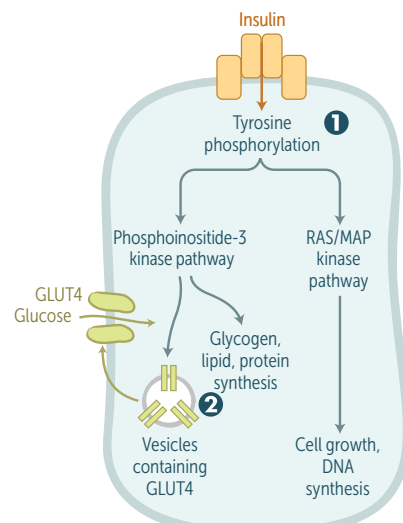
Brain utilizes glucose for metabolism normally and ketone bodies during starvation. RBCs utilize glucose because they lack mitochondria for aerobic metabolism.

BRICK L (insulin-independent glucose uptake):
Brain, RBCs, Intestine, Cornea, Kidney, Liver.

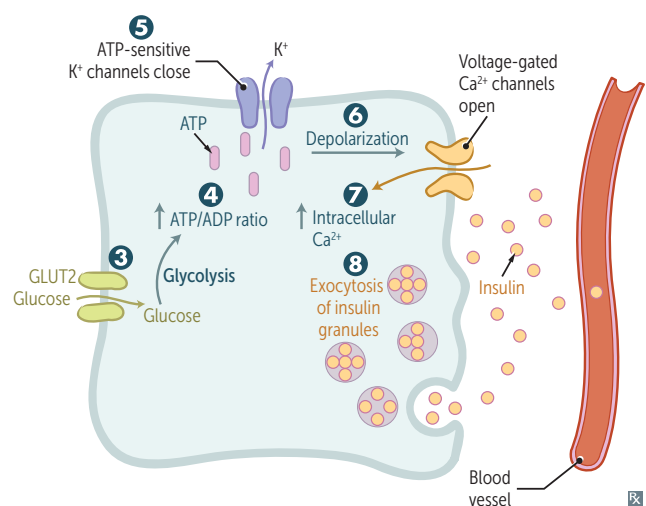
REGULATION

Glucose is the major regulator of insulin release. ↑ insulin response with oral vs IV glucose because of incretins such as glucagon-like peptide 1 (GLP-1) and glucose-dependent insulinotropic polypeptide (GIP), which are released after meals and ↑ β cell sensitivity to glucose. Release ↓ by α₂, ↑ by β₂ (2 = regulates **insulin**)

Glucose enters β cells **3** → ↑ ATP generated from glucose metabolism **4** closes K⁺ channels (target of sulfonylureas) **5** and depolarizes β cell membrane **6**. Voltage-gated Ca²⁺ channels open → Ca²⁺ influx **7** and stimulation of insulin exocytosis **8**.



Insulin-dependent glucose uptake



Insulin secretion by pancreatic β cells

Glucagon

SOURCE	Made by α cells of pancreas.
FUNCTION	Glycogenolysis, gluconeogenesis, lipolysis, ketone production.
REGULATION	Secreted in response to hypoglycemia. Inhibited by insulin, hyperglycemia, and somatostatin.

Hypothalamic-pituitary hormones

HORMONE	FUNCTION	CLINICAL NOTES
CRH	↑ ACTH, MSH, β -endorphin	↓ in chronic exogenous steroid use.
Dopamine	↓ prolactin, TSH	Dopamine antagonists (eg, antipsychotics) can cause galactorrhea due to hyperprolactinemia.
GHRH	↑ GH	Analog (tesamorelin) used to treat HIV-associated lipodystrophy.
GnRH	↑ FSH, LH	Suppressed by hyperprolactinemia. Tonic GnRH suppresses HPG axis. Pulsatile GnRH leads to puberty, fertility.
Prolactin	↓ GnRH	Pituitary prolactinoma → amenorrhea, osteoporosis, hypogonadism, galactorrhea.
Somatostatin	↓ GH, TSH	Analog used to treat acromegaly.
TRH	↑ TSH, prolactin	↑ TRH (eg, in 1°/2° hypothyroidism) may increase prolactin secretion → galactorrhea.

Prolactin**SOURCE**

Secreted mainly by anterior pituitary.

Structurally homologous to growth hormone.

FUNCTION

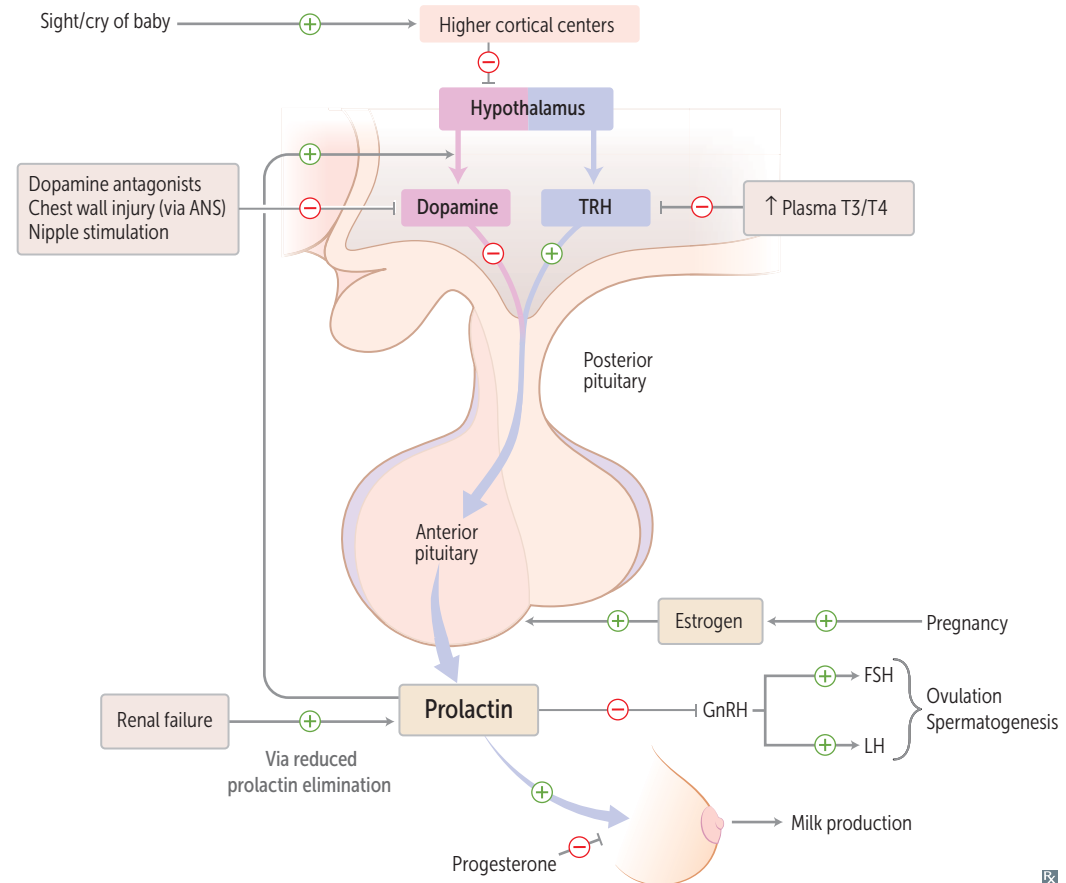
Stimulates milk production in breast; inhibits ovulation in females and spermatogenesis in males by inhibiting GnRH synthesis and release.

Excessive amounts of prolactin associated with ↓ libido.

REGULATION

Prolactin secretion from anterior pituitary is tonically inhibited by dopamine from tuberoinfundibular pathway of hypothalamus. Prolactin in turn inhibits its own secretion by ↑ dopamine synthesis and secretion from hypothalamus. TRH ↑ prolactin secretion (eg, in 1° or 2° hypothyroidism).

Dopamine agonists (eg, bromocriptine) inhibit prolactin secretion and can be used in treatment of prolactinoma. Dopamine antagonists (eg, most antipsychotics) and estrogens (eg, OCPs, pregnancy) stimulate prolactin secretion.



Growth hormone (somatotropin)

SOURCE	Secreted by anterior pituitary.	
FUNCTION	Stimulates linear growth and muscle mass through IGF-1 (somatomedin C) secretion by liver. ↑ insulin resistance (diabetogenic).	Somatostatin keeps your growth static. Somatomedin mediates your growth.
REGULATION	Released in pulses in response to growth hormone–releasing hormone (GHRH). Secretion ↑ during exercise, deep sleep, puberty, hypoglycemia. Secretion inhibited by glucose and somatostatin release via negative feedback by somatomedin.	Excess secretion of GH (eg, pituitary adenoma) may cause acromegaly (adults) or gigantism (children). Treat with somatostatin analogs (eg, octreotide) or surgery.

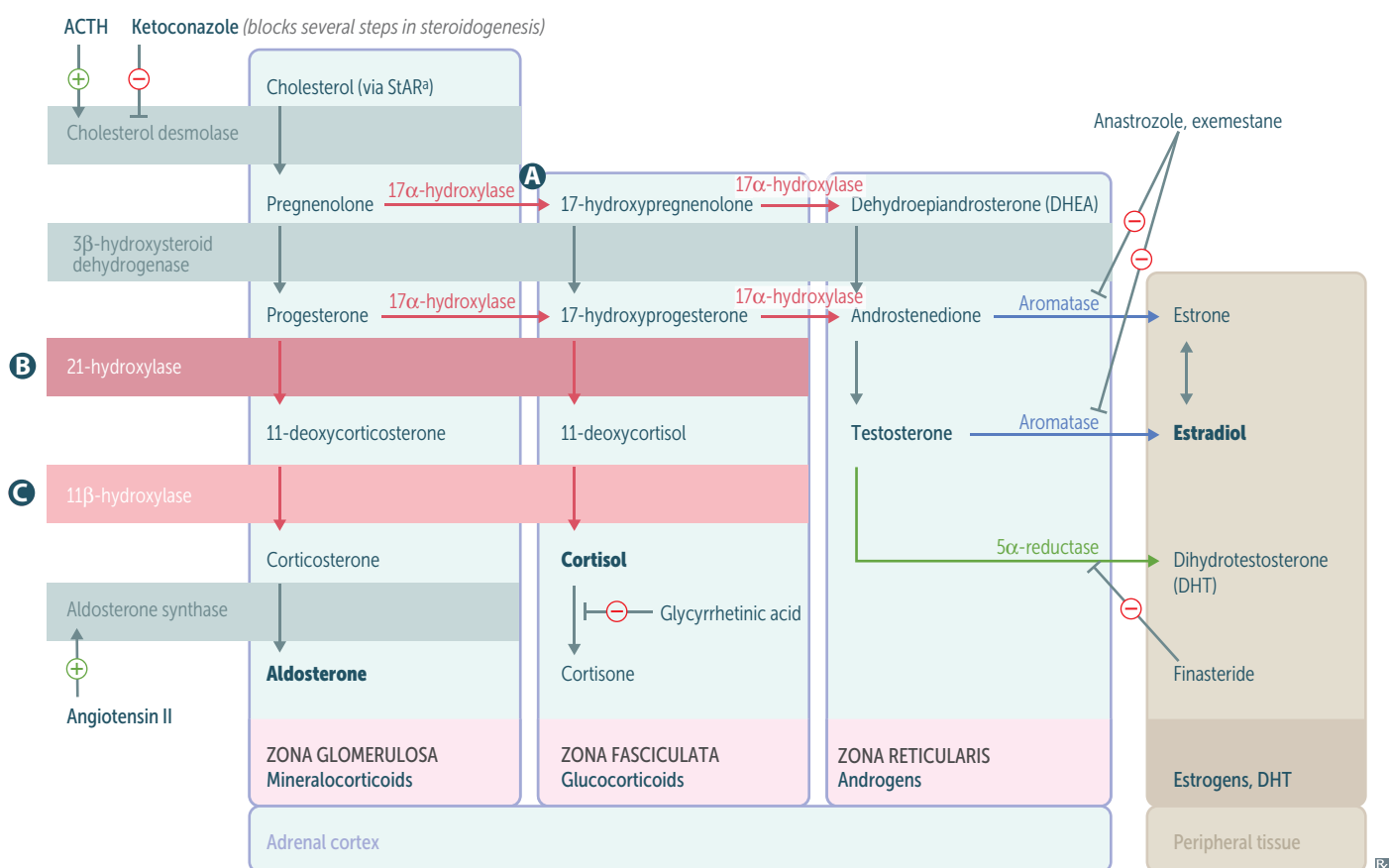
Appetite regulation

Ghrelin	Stimulates hunger (orexigenic effect) and GH release (via GH secretagog receptor). Produced by stomach. Sleep deprivation or Prader-Willi syndrome → ↑ ghrelin production.	Ghrelin makes you hunghre.
Leptin	Satiety hormone. Produced by adipose tissue. Mutation of leptin gene → congenital obesity. Sleep deprivation or starvation → ↓ leptin production.	Leptin keeps you thin.
Endocannabinoids	Act at cannabinoid receptors in hypothalamus and nucleus accumbens, two key brain areas for the homeostatic and hedonic control of food intake → ↑ appetite.	Exogenous cannabinoids cause “the munchies.”

Antidiuretic hormone

SOURCE	Synthesized in hypothalamus (supraoptic nuclei), stored and secreted by posterior pituitary.	
FUNCTION	Regulates serum osmolarity (V_2 -receptors) and blood pressure (V_1 -receptors). Primary function is serum osmolarity regulation (ADH ↓ serum osmolarity, ↑ urine osmolarity) via regulation of aquaporin channel insertion in principal cells of renal collecting duct.	ADH level is ↓ in central diabetes insipidus (DI), normal or ↑ in nephrogenic DI. Nephrogenic DI can be caused by mutation in V_2 -receptor. Desmopressin acetate (ADH analog) is a treatment for central DI and nocturnal enuresis.
REGULATION	Osmoreceptors in hypothalamus (1°); hypovolemia.	

Adrenal steroids and congenital adrenal hyperplasias

^aRate-limiting step.

ENZYME DEFICIENCY	MINERALOCORTICOIDS	CORTISOL	SEX HORMONES	BP	[K ⁺]	LABS	PRESENTATION
A 17α-hydroxylase^a	↑	↓	↓	↑	↓	↓ androstenedione	XY: ambiguous genitalia, undescended testes XX: lacks 2° sexual development
B 21-hydroxylase^a	↓	↓	↑	↓	↑	↑ renin activity ↑ 17-hydroxyprogesterone	Most common Presents in infancy (salt wasting) or childhood (precocious puberty) XX: virilization
C 11β-hydroxylase^a	↓ aldosterone ↑ 11-deoxycorticosterone (results in ↑ BP)	↓	↑	↑	↓	↓ renin activity	XX: virilization

^aAll congenital adrenal enzyme deficiencies are characterized by an enlargement of both adrenal glands due to ↑ ACTH stimulation (in response to ↓ cortisol) and by skin hyperpigmentation.

Cortisol

SOURCE	Adrenal zona fasciculata.	Bound to corticosteroid-binding globulin.
FUNCTION	↑ Appetite ↑ Blood pressure: - Upregulates α_1-receptors on arterioles → ↑ sensitivity to norepinephrine and epinephrine (permissive action) - At high concentrations, can bind to mineralocorticoid (aldosterone) receptors ↑ Insulin resistance (diabetogenic) ↑ Gluconeogenesis, lipolysis, and proteolysis (↓ glucose utilization) ↓ Fibroblast activity (poor wound healing, ↓ collagen synthesis, ↑ striae) ↓ Inflammatory and Immune responses: - Inhibits production of leukotrienes and prostaglandins - Inhibits WBC adhesion → neutrophilia - Blocks histamine release from mast cells - Eosinopenia, lymphopenia - Blocks IL-2 production ↓ Bone formation (↓ osteoblast activity)	Cortisol is a A BIG FIB. Exogenous corticosteroids can cause reactivation of TB and candidiasis (blocks IL-2 production).
REGULATION	CRH (hypothalamus) stimulates ACTH release (pituitary) → cortisol production in adrenal zona fasciculata. Excess cortisol ↓ CRH, ACTH, and cortisol secretion.	Chronic stress induces prolonged secretion.

Calcium homeostasis

Plasma Ca^{2+} exists in three forms:	- Ionized/free (~ 45%, active form) - Bound to albumin (~ 40%) - Bound to anions (~ 15%)	↑ in pH → ↑ affinity of albumin (↑ negative charge) to bind Ca^{2+} → hypocalcemia (cramps, pain, paresthesias, carpopedal spasm). Ionized/free Ca^{2+} is 1° regulator of PTH; changes in pH alter PTH secretion, whereas changes in albumin do not.
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Vitamin D

SOURCE	D_3 from exposure of skin to sun, ingestion of fish and plants. D_2 from ingestion of plants, fungi, yeasts. Both converted to 25-OH in liver and to 1,25-(OH) $_2$ vitamin D (active form) in kidney.	Deficiency → rickets in kids, osteomalacia in adults. Caused by malabsorption, ↓ sunlight, poor diet, chronic kidney failure.
FUNCTION	↑ absorption of dietary Ca^{2+} and PO_4^{3-} . Enhances bone mineralization.	24,25-(OH) $_2$ D_3 is an inactive form of vitamin D. PTH leads to ↑ Ca^{2+} reabsorption and ↓ PO_4^{3-} reabsorption in the kidney, whereas 1,25-(OH) $_2$ D_3 leads to ↑ absorption of both Ca^{2+} and PO_4^{3-} in the gut.
REGULATION	↑ PTH, ↓ Ca^{2+}, ↓ PO_4^{3-} → ↑ 1,25-(OH)$_2$ production. 1,25-(OH)$_2$ feedback inhibits its own production.	

Parathyroid hormone

SOURCE

Chief cells of parathyroid.

FUNCTION

↑ bone resorption of Ca^{2+} and PO_4^{3-} .
 ↑ kidney reabsorption of Ca^{2+} in distal convoluted tubule.
 ↓ reabsorption of PO_4^{3-} in proximal convoluted tubule.
 ↑ $1,25\text{-(OH)}_2\text{D}_3$ (calcitriol) production by stimulating kidney 1α -hydroxylase in proximal convoluted tubule.

PTH ↑ serum Ca^{2+} , ↓ serum $\text{(PO}_4^{3-}\text{)}$, ↑ urine $\text{(PO}_4^{3-}\text{)}$, ↑ urine cAMP.

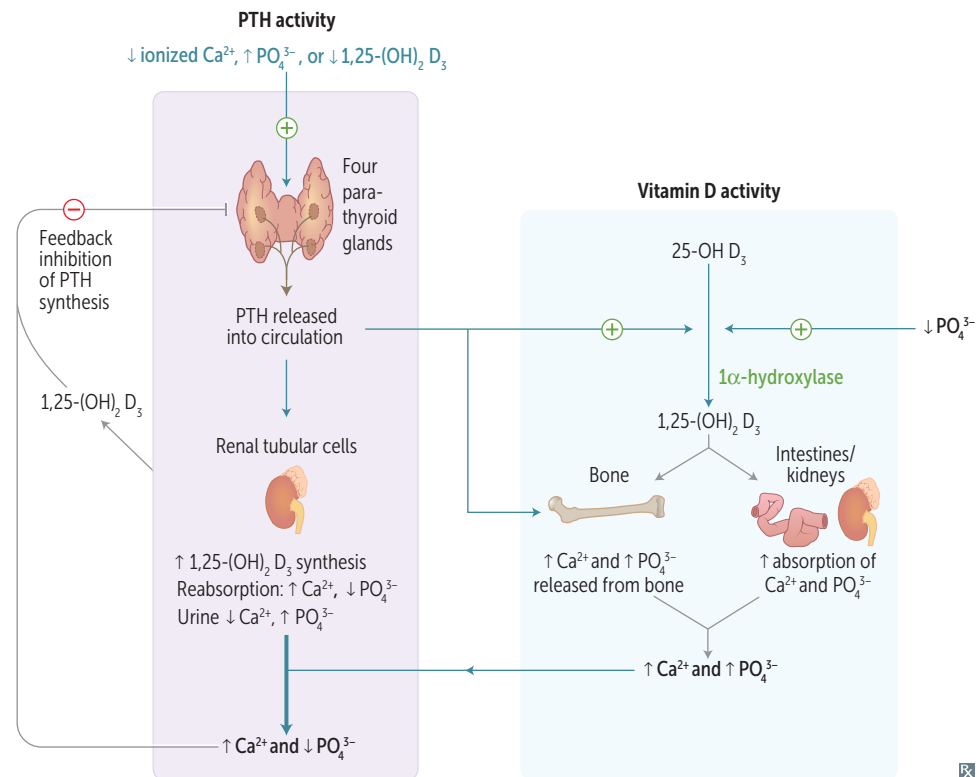
↑ RANK-L (receptor activator of NF- κ B ligand) secreted by osteoblasts and osteocytes. Binds RANK (receptor) on osteoclasts and their precursors to stimulate osteoclasts and ↑ Ca^{2+} → bone resorption. Intermittent PTH release can also stimulate bone formation.

PTH = Phosphate-Trashing Hormone.

PTH-related peptide (PTHrP) functions like PTH and is commonly increased in malignancies (eg, squamous cell carcinoma of the lung, renal cell carcinoma).

REGULATION

↓ serum Ca^{2+} → ↑ PTH secretion.
 ↑ serum PO_4^{3-} → ↑ PTH secretion.
 ↓ serum Mg^{2+} → ↑ PTH secretion.
 ↓↓ serum Mg^{2+} → ↓ PTH secretion.
 Common causes of ↓ Mg^{2+} include diarrhea, aminoglycosides, diuretics, alcohol abuse.



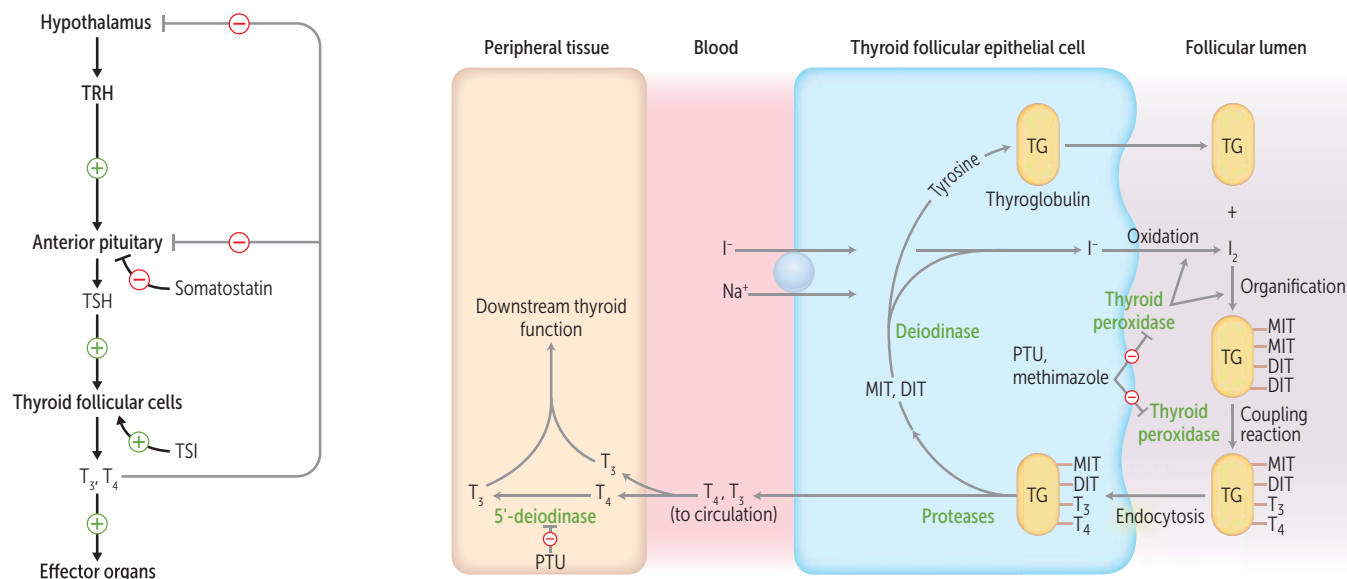
Calcitonin

SOURCE	Parafollicular cells (C cells) of thyroid.	Calcitonin opposes actions of PTH. Not important in normal Ca^{2+} homeostasis.
FUNCTION	$\downarrow$ bone resorption of Ca^{2+} .	Calcitonin tones down serum Ca^{2+} levels and keeps it in bones .
REGULATION	$\uparrow$ serum $\text{Ca}^{2+} \rightarrow$ calcitonin secretion.	

Thyroid hormones (T_3/T_4)

Iodine-containing hormones that control the body's metabolic rate.

SOURCE	Follicles of thyroid. Most T_3 formed in target tissues.	T_3 functions— 4 B's : B rain maturation B one growth β -adrenergic effects B asal metabolic rate $\uparrow$
FUNCTION	Bone growth (synergism with GH) CNS maturation $\uparrow \beta_1$ receptors in heart = $\uparrow$ CO, HR, SV, contractility $\uparrow$ basal metabolic rate via $\uparrow \text{Na}^+/\text{K}^+$ -ATPase activity $\rightarrow \uparrow \text{O}_2$ consumption, RR, body temperature $\uparrow$ glycogenolysis, gluconeogenesis, lipolysis	Thyroxine-binding globulin (TBG) binds most T_3/T_4 in blood; only free hormone is active. $\downarrow$ TBG in hepatic failure, steroids; $\uparrow$ TBG in pregnancy or OCP use (estrogen $\uparrow$ TBG). T_4 is major thyroid product; converted to T_3 in peripheral tissue by 5'-deiodinase. T_3 binds nuclear receptor with greater affinity than T_4 . Thyroid peroxidase is the enzyme responsible for oxidation and organification of iodide as well as coupling of monoiodotyrosine (MIT) and di-iodotyrosine (DIT). $\text{DIT} + \text{DIT} = \text{T}_4$. $\text{DIT} + \text{MIT} = \text{T}_3$. Propylthiouracil (PTU) inhibits both thyroid peroxidase and 5'-deiodinase. Methimazole inhibits thyroid peroxidase only. Glucocorticoids inhibit peripheral conversion of T_4 to T_3 .
REGULATION	TRH (hypothalamus) stimulates TSH (pituitary), which stimulates follicular cells. May also be stimulated by thyroid-stimulating immunoglobulin (TSI) in Graves disease. Negative feedback primarily by free T_3/T_4 to anterior pituitary ($\downarrow$ sensitivity to TRH) and hypothalamus ($\downarrow$ TRH secretion). Wolff-Chaikoff effect—excess iodine temporarily inhibits thyroid peroxidase $\rightarrow \downarrow$ iodine organification $\rightarrow \downarrow \text{T}_3/\text{T}_4$ production.	



Signaling pathways of endocrine hormones

cAMP	F SH, L H, A CTH, T SH, C RH, h CG, A DH (V ₂ -receptor), M SH, P TH, calcitonin, GHRH, glucagon, histamine (H ₂ -receptor)	FLAT ChAMP
cGMP	B NP, A NP, E DRF (NO)	BAD GraMPa Think vasodilators
IP₃	G nRH, O xytocin, A DH (V ₁ -receptor), T RH, H istamine (H ₁ -receptor), A ngiotensin II, G astrin	GOAT HAG
Intracellular receptor	P rogestosterone, E strogen, T estosterone, C ortisol, A ldosterone, T ₃ / T ₄ , V itamin D	PET CAT on TV
Receptor tyrosine kinase	Insulin, I GF-1, F GF, P DGF, E GF	MAP kinase pathway Think G rowth F actors
Nonreceptor tyrosine kinase	P rolactin, I mmunomodulators (eg, cytokines IL-2, IL-6, IFN), G H, G -CSF, E rythropoietin, T hrombopoietin	JAK/STAT pathway Think acidophils and cytokines PIGGLET

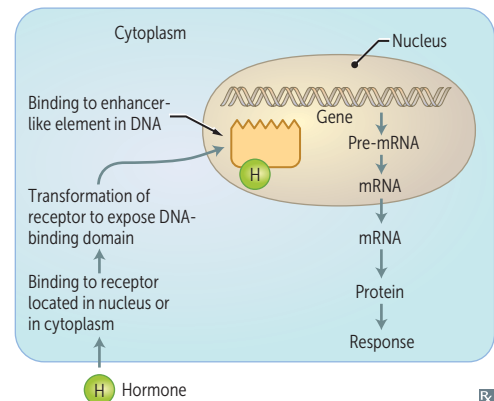
Signaling pathway of steroid hormones

Steroid hormones are lipophilic and therefore must circulate bound to specific binding globulins, which ↑ their solubility.

In men, ↑ sex hormone-binding globulin (SHBG) lowers free testosterone
→ gynecomastia.

In women, ↓ SHBG raises free testosterone
→ hirsutism.

OCPs, pregnancy → ↑ SHBG.



► ENDOCRINE—PATHOLOGY

Cushing syndrome

ETIOLOGY

↑ cortisol due to a variety of causes:

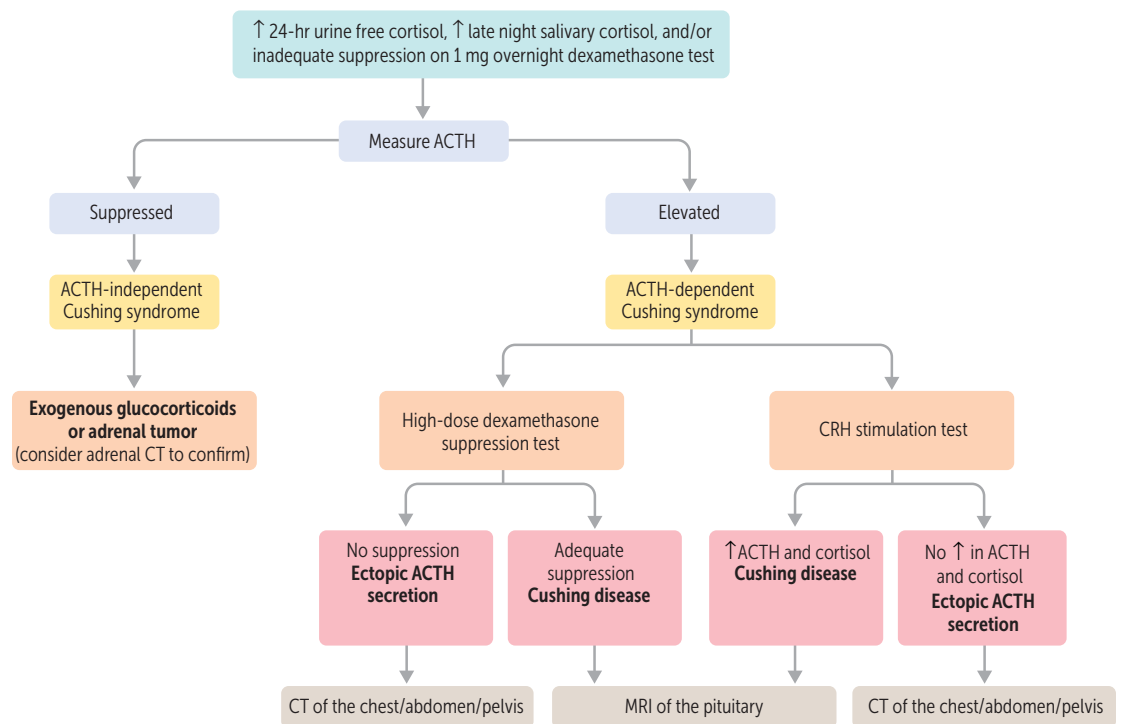
- Exogenous corticosteroids—result in ↓ ACTH, bilateral adrenal atrophy. Most common cause.
- Primary adrenal adenoma, hyperplasia, or carcinoma—result in ↓ ACTH, atrophy of uninvolved adrenal gland. Can also present with pseudohyperaldosteronism.
- ACTH-secreting pituitary adenoma (Cushing disease); paraneoplastic ACTH secretion (eg, small cell lung cancer, bronchial carcinoids)—result in ↑ ACTH, bilateral adrenal hyperplasia. Cushing disease is responsible for the majority of endogenous cases of Cushing syndrome.

FINDINGS

Hypertension, weight gain, moon facies **A**, abdominal striae **B** and truncal obesity, buffalo hump, skin changes (eg, thinning, striae), osteoporosis, hyperglycemia (insulin resistance), amenorrhea, immunosuppression.

DIAGNOSIS

Screening tests include: ↑ free cortisol on 24-hr urinalysis, ↑ midnight salivary cortisol, and no suppression with overnight low-dose dexamethasone test. Measure serum ACTH. If ↓, suspect adrenal tumor or exogenous glucocorticoids. If ↑, distinguish between Cushing disease and ectopic ACTH secretion (eg, from small cell lung cancer) with a high-dose dexamethasone suppression test and CRH stimulation test. Ectopic secretion will not decrease with dexamethasone because the source is resistant to negative feedback; ectopic secretion will not increase with CRH because pituitary ACTH is suppressed.

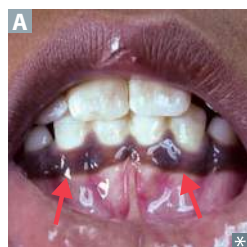


Adrenal insufficiency

Inability of adrenal glands to generate enough glucocorticoids +/- mineralocorticoids for the body's needs. Symptoms include weakness, fatigue, orthostatic hypotension, muscle aches, weight loss, GI disturbances, sugar and/or salt cravings. Treatment: glucocorticoid/mineralocorticoid replacement.

Diagnosis involves measurement of serum electrolytes, morning/random serum cortisol and ACTH (low cortisol, high ACTH in 1° adrenal insufficiency; low cortisol, low ACTH in 2°/3° adrenal insufficiency due to pituitary/hypothalamic disease), and response to ACTH stimulation test.

Alternatively, can use metyrapone stimulation test: metyrapone blocks last step of cortisol synthesis (11-deoxycortisol → cortisol). Normal response is ↓ cortisol and compensatory ↑ ACTH and 11-deoxycortisol. In 1° adrenal insufficiency, ACTH is ↑ but 11-deoxycortisol remains ↓ after test. In 2°/3° adrenal insufficiency, both ACTH and 11-deoxycortisol remain ↓ after test.

Primary adrenal insufficiency

Deficiency of aldosterone and cortisol production due to loss of gland function → hypotension (hyponatremic volume contraction), hyperkalemia, metabolic acidosis, skin and mucosal hyperpigmentation **A** (due to ↑ MSH, a byproduct of ACTH production from proopiomelanocortin [POMC]).

- **Acute**—sudden onset (eg, due to massive hemorrhage). May present with shock in acute adrenal crisis.
- **Chronic**—aka **Addison disease**. Due to adrenal atrophy or destruction by disease (autoimmune destruction most common in the Western world; TB most common in the developing world).

Primary Pigments the skin/mucosa.

Associated with autoimmune polyglandular syndromes.

Waterhouse-Friderichsen syndrome—acute 1° adrenal insufficiency due to adrenal hemorrhage associated with septicemia (usually *Neisseria meningitidis*), DIC, endotoxic shock.

Secondary adrenal insufficiency

Seen with ↓ pituitary ACTH production. No skin/mucosal hyperpigmentation, no hyperkalemia (aldosterone synthesis preserved due to intact renin-angiotensin-aldosterone axis).

Secondary Spares the skin/mucosa.

Tertiary adrenal insufficiency

Seen in patients with chronic exogenous steroid use, precipitated by abrupt withdrawal. Aldosterone synthesis unaffected.

Tertiary from **T**reatment.

Hyperaldosteronism

Increased secretion of aldosterone from adrenal gland. Clinical features include hypertension, ↓ or normal K⁺, metabolic alkalosis. No edema due to aldosterone escape mechanism.

Primary hyperaldosteronism

Seen with adrenal adenoma (Conn syndrome) or bilateral adrenal hyperplasia. ↑ aldosterone, ↓ renin.

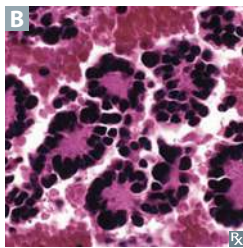
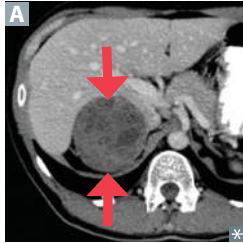
Secondary hyperaldosteronism

Seen in patients with renovascular hypertension, juxtaglomerular cell tumors (renin-producing), and edema (eg, cirrhosis, heart failure, nephrotic syndrome).

Neuroendocrine tumors

Group of neoplasms originating from Kulchitsky and enterochromaffin-like cells. Occur in various organs (eg, thyroid: medullary carcinoma; lungs: small cell carcinoma; pancreas: islet cell tumor; adrenals: pheochromocytoma). Cells contain amine precursor uptake decarboxylase (APUD) and secrete different hormones (eg, 5-HIAA, neuron-specific enolase [NSE], chromogranin A).

Neuroblastoma



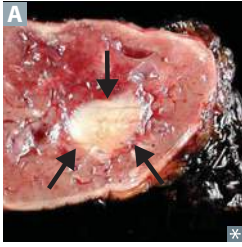
Most common tumor of the adrenal medulla **A** in **children**, usually < 4 years old. Originates from neural crest cells. Occurs anywhere along the sympathetic chain.

Most common presentation is abdominal distension and a firm, irregular mass that can cross the midline (vs Wilms tumor, which is smooth and unilateral). Less likely to develop hypertension than with pheochromocytoma. Can also present with opsoclonus-myoclonus syndrome (“dancing eyes-dancing feet”).

↑ HVA and VMA (catecholamine metabolites) in urine. Homer-Wright rosettes **B** characteristic of neuroblastoma and medulloblastoma. Bombesin and NSE ⊕. Associated with overexpression of *N-myc* oncogene. Classified as an APUD tumor.

Pheochromocytoma

ETIOLOGY



Most common tumor of the adrenal medulla in **adults** **A**. Derived from chromaffin cells (arise from neural crest).
May be associated with germline mutations (eg, *NF-1*, *VHL*, *RET* [MEN 2A, 2B]).

Rule of 10's:

- 10%** malignant
- 10%** bilateral
- 10%** extra-adrenal (eg, bladder wall, organ of Zuckerkandl)
- 10%** calcify
- 10%** kids

SYMPTOMS

Most tumors secrete epinephrine, norepinephrine, and dopamine, which can cause episodic hypertension. Symptoms occur in “spells”—relapse and remit.

Episodic hyperadrenergic symptoms (**5 P's**):

- P**ressure (↑ BP)
- P**ain (headache)
- P**erspiration
- P**alpitations (tachycardia)
- P**allor

FINDINGS

↑ catecholamines and metanephrines in urine and plasma.

TREATMENT

Irreversible α -antagonists (eg, phenoxybenzamine) followed by β -blockers prior to tumor resection. α -blockade must be achieved before giving β -blockers to avoid a hypertensive crisis.

Phenoxybenzamine (16 letters) is given for **pheochromocytoma** (also 16 letters).

Hypothyroidism vs hyperthyroidism

	Hypothyroidism	Hyperthyroidism
SIGNS/SYMPTOMS	Cold intolerance (↓ heat production)	Heat intolerance (↑ heat production)
	Weight gain, ↓ appetite	Weight loss, ↑ appetite
	Hypoactivity, lethargy, fatigue, weakness, depressed mood	Hyperactivity, anxiety, insomnia, hand tremor
	Constipation	Diarrhea/hyperdefecation
	↓ reflexes (delayed/slow relaxing)	↑ reflexes (brisk)
	Hypothyroid myopathy (proximal muscle weakness, ↑ CK)	Thyrotoxic myopathy (proximal muscle weakness, normal CK)
	Myxedema (facial/periorbital)	Peritibial myxedema (Graves disease), periorbital edema
	Dry, cool skin; coarse, brittle hair	Warm, moist skin; fine hair
LAB FINDINGS	Bradycardia, dyspnea on exertion	Chest pain, palpitations, and arrhythmias (eg, atrial fibrillation) due to ↑ number and sensitivity of β-adrenergic receptors
	↑ TSH (if 1°)	↓ TSH (if 1°)
	↓ free T ₃ and T ₄	↑ free or total T ₃ and T ₄
	Hypercholesterolemia (due to ↓ LDL receptor expression)	Hypocholesterolemia (due to ↑ LDL receptor expression)

Causes of goiter

	Smooth/diffuse	Nodular
	Graves disease	Toxic multinodular goiter
	Hashimoto thyroiditis	Thyroid adenoma
	Iodine deficiency	Thyroid cancer
	TSH-secreting pituitary adenoma	Thyroid cyst

Hypothyroidism

Hashimoto thyroiditis

Most common cause of hypothyroidism in iodine-sufficient regions; an autoimmune disorder with antithyroid peroxidase (antimicrosomal) and antithyroglobulin antibodies. Associated with ↑ risk of non-Hodgkin lymphoma (typically of B-cell origin).

May be hyperthyroid early in course due to thyrotoxicosis during follicular rupture.

Histologic findings: Hürthle cells, lymphoid aggregates with germinal centers **A**.

Findings: moderately enlarged, **nontender** thyroid.

Congenital hypothyroidism (cretinism)

Severe fetal hypothyroidism due to maternal hypothyroidism, thyroid agenesis, thyroid dysgenesis (most common cause in US), iodine deficiency, dysmorphogenetic goiter.

Findings: **P**ot-bellied, **P**ale, **P**uffy-faced child with **P**rotruding umbilicus, **P**rotuberant tongue, and **P**oor brain development: the **6 P's** **B** **C**.

Subacute granulomatous thyroiditis (de Quervain)

Self-limited disease often following a flu-like illness (eg, viral infection).

May be hyperthyroid early in course, followed by hypothyroidism.

Histology: granulomatous inflammation.

Findings: ↑ ESR, jaw pain, very **tender** thyroid. (de **Quervain** is associated with **pain**.)

Riedel thyroiditis

Thyroid replaced by fibrous tissue with inflammatory infiltrate **D**. Fibrosis may extend to local structures (eg, trachea, esophagus), mimicking anaplastic carcinoma. 1/3 are hypothyroid.

Considered a manifestation of IgG₄-related systemic disease (eg, autoimmune pancreatitis, retroperitoneal fibrosis, noninfectious aortitis).

Findings: fixed, hard (rock-like), **painless** goiter.

Other causes

Iodine deficiency **E**, goitrogens (eg, amiodarone, lithium), Wolff-Chaikoff effect (thyroid gland downregulation in response to ↑ iodide).



Hyperthyroidism**Graves disease**

Most common cause of hyperthyroidism. Thyroid-stimulating immunoglobulin (IgG; type II hypersensitivity) stimulates TSH receptors on thyroid (hyperthyroidism, diffuse goiter) and dermal fibroblasts (pretibial myxedema). Infiltration of retroorbital space by activated T-cells → ↑ cytokines (eg, TNF- α , IFN- γ) → ↑ fibroblast secretion of hydrophilic GAGs → ↑ osmotic muscle swelling, muscle inflammation, and adipocyte count → exophthalmos **A**. Often presents during stress (eg, pregnancy). Associated with HLA-DR3 and HLA-B8. Tall, crowded follicular epithelial cells; scalloped colloid **B**.

Toxic multinodular goiter

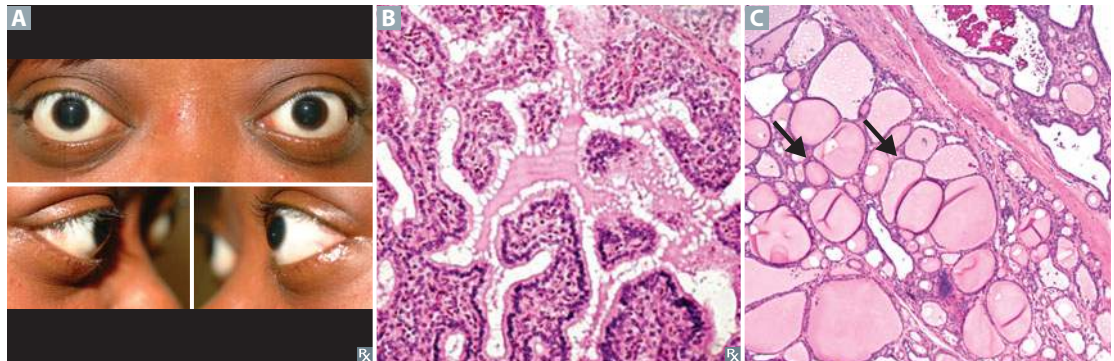
Focal patches of hyperfunctioning follicular cells distended with colloid **C** working independently of TSH (due to TSH receptor mutations in 60% of cases). ↑ release of T₃ and T₄. Hot nodules are rarely malignant.

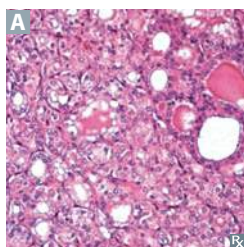
Thyroid storm

Uncommon but serious complication that occurs when hyperthyroidism is incompletely treated/untreated and then significantly worsens in the setting of acute stress such as infection, trauma, surgery. Presents with agitation, delirium, fever, diarrhea, coma, and tachyarrhythmia (cause of death). May see ↑ LFTs. Treat with the **4 P's**: β -blockers (eg, **P**ropranolol), **P**ropylthiouracil, corticosteroids (eg, **P**rednisolone), **P**otassium iodide (Lugol iodine).

Jod-Basedow phenomenon

Thyrotoxicosis if a patient with iodine deficiency and partially autonomous thyroid tissue (eg, autonomous nodule) is made iodine replete. Opposite of Wolff-Chaikoff effect.

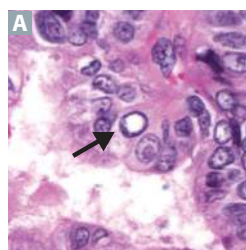


Thyroid adenoma

Benign solitary growth of the thyroid. Most are nonfunctional (“cold”), can rarely cause hyperthyroidism via autonomous thyroid hormone production (“hot” or “toxic”). Most common histology is follicular **A**; absence of capsular or vascular invasion (unlike follicular carcinoma).

Thyroid cancer

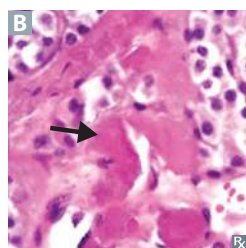
Typically diagnosed with fine needle aspiration; treated with thyroidectomy. Complications of surgery include hoarseness (due to recurrent laryngeal nerve damage), hypocalcemia (due to removal of parathyroid glands), and transection of recurrent and superior laryngeal nerves (during ligation of inferior thyroid artery and superior laryngeal artery, respectively).

Papillary carcinoma

Most common, excellent prognosis. Empty-appearing nuclei with central clearing (“**Orphan Annie**” eyes) **A**, psam**Moma** bodies, nuclear grooves (**Papi** and **Moma** adopted **Orphan Annie**). ↑ risk with *RET* and *BRAF* mutations, childhood irradiation.

Follicular carcinoma

Good prognosis. Invades thyroid capsule and vasculature (unlike follicular adenoma), uniform follicles; hematogenous spread is common. Associated with *RAS* mutation.

Medullary carcinoma

From parafollicular “C cells”; produces calcitonin, sheets of cells in an amyloid stroma (stains with Congo red **B**). Associated with MEN 2A and 2B (*RET* mutations).

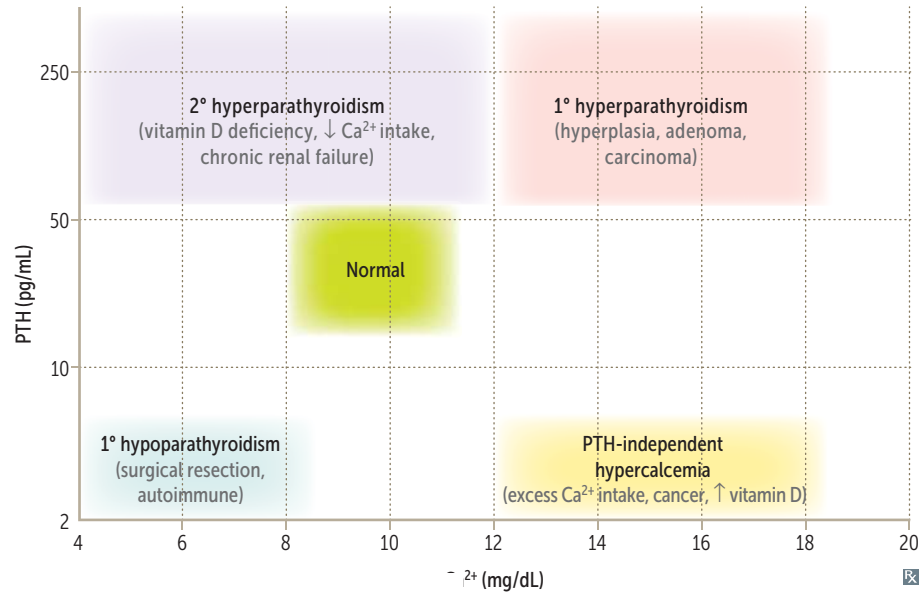
**Undifferentiated/
anaplastic carcinoma**

Older patients; invades local structures, very poor prognosis.

Lymphoma

Associated with Hashimoto thyroiditis.

Diagnosis of parathyroid disease



Hypoparathyroidism

Due to accidental surgical excision of parathyroid glands, autoimmune destruction, or DiGeorge syndrome. Findings: tetany, hypocalcemia, hyperphosphatemia.

Chvostek sign—tapping of facial nerve (tap the **C**heek) → contraction of facial muscles.

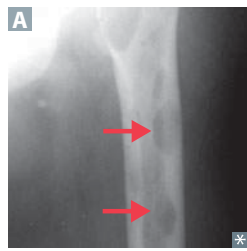
Trousseau sign—occlusion of brachial artery with BP cuff (cuff the **T**riceps) → carpal spasm.

Pseudohypoparathyroidism type 1A (Albright hereditary osteodystrophy)—unresponsiveness of kidney to PTH → hypocalcemia despite ↑ PTH levels. Characterized by shortened 4th/5th digits, short stature. Autosomal dominant. Due to defective G_s protein α-subunit causing end-organ resistance to PTH. Defect must be inherited from mother due to imprinting.

Pseudopseudohypoparathyroidism—physical exam features of Albright hereditary osteodystrophy but without end-organ PTH resistance (PTH level normal). Occurs when defective G_s protein α-subunit is inherited from father.

Hyperparathyroidism

Primary hyperparathyroidism



Usually due to parathyroid adenoma or hyperplasia. **Hypercalcemia**, hypercalciuria (renal **stones**), polyuria (**thrones**), hypophosphatemia, $\uparrow$ PTH, $\uparrow$ ALP, $\uparrow$ cAMP in urine. Most often asymptomatic. May present with weakness and constipation (**“groans”**), abdominal/flank pain (kidney stones, acute pancreatitis), depression (**“psychiatric overtones”**).

Osteitis fibrosa cystica—cystic **bone** spaces filled with brown fibrous tissue **A** (“brown tumor” consisting of osteoclasts and deposited hemosiderin from hemorrhages; causes bone pain). Due to $\uparrow$ PTH, classically associated with 1° (but also seen with 2°) hyperparathyroidism.

“Stones, thrones, bones, groans, and psychiatric overtones.”

Secondary hyperparathyroidism

2° hyperplasia due to $\downarrow$ Ca^{2+} absorption and/or $\uparrow$ PO_4^{3-} , most often in chronic renal disease (causes hypovitaminosis D and hyperphosphatemia $\rightarrow$ $\downarrow$ Ca^{2+}).

Hypocalcemia, hyperphosphatemia in chronic renal failure (vs hypophosphatemia with most other causes), $\uparrow$ ALP, $\uparrow$ PTH.

Renal osteodystrophy—renal disease $\rightarrow$ 2° and 3° hyperparathyroidism $\rightarrow$ bone lesions.

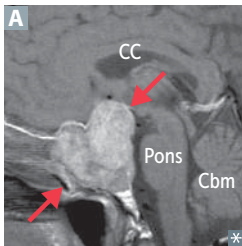
Tertiary hyperparathyroidism

Refractory (autonomous) hyperparathyroidism resulting from chronic renal disease. $\uparrow\uparrow$ PTH, $\uparrow$ Ca^{2+} .

Familial hypocalciuric hypercalcemia

Defective G-coupled Ca^{2+} -sensing receptors in multiple tissues (eg, parathyroids, kidneys). Higher than normal Ca^{2+} levels required to suppress PTH. Excessive renal Ca^{2+} reuptake $\rightarrow$ mild hypercalcemia and hypocalciuria with normal to $\uparrow$ PTH levels.

Pituitary adenoma



Benign tumor, most commonly prolactinoma (arises from lactotrophs). Adenoma **A** may be functional (hormone producing) or nonfunctional (silent). Nonfunctional tumors present with mass effect (bitemporal hemianopia, hypopituitarism, headache). Functional tumor presentation is based on the hormone produced.

Prolactinoma in women classically presents as galactorrhea, amenorrhea, and $\downarrow$ bone density due to suppression of estrogen. Prolactinoma in men classically presents as low libido and infertility.

Treatment: dopamine agonists (eg, bromocriptine, cabergoline), transsphenoidal resection.

Nelson syndrome

Enlargement of existing ACTH-secreting pituitary adenoma after bilateral adrenalectomy for refractory Cushing disease (due to removal of cortisol feedback mechanism). Presents with hyperpigmentation, headaches and bitemporal hemianopia. Treatment: pituitary irradiation or surgical resection.

Acromegaly

Excess GH in adults. Typically caused by pituitary adenoma.

FINDINGS	Large tongue with deep furrows, deep voice, large hands and feet, coarsening of facial features with aging A , frontal bossing, diaphoresis (excessive sweating), impaired glucose tolerance (insulin resistance). ↑ risk of colorectal polyps and cancer.	↑ GH in children → gigantism (↑ linear bone growth). HF most common cause of death.
DIAGNOSIS	↑ serum IGF-1; failure to suppress serum GH following oral glucose tolerance test; pituitary mass seen on brain MRI.	
TREATMENT	Pituitary adenoma resection. If not cured, treat with octreotide (somatostatin analog) or pegvisomant (growth hormone receptor antagonist), dopamine agonists (eg, cabergoline).	

**Laron syndrome (dwarfism)**

Defective growth hormone receptors → ↓ linear growth. ↑ GH, ↓ IGF-1. Clinical features include short height, small head circumference, characteristic facies with saddle nose and prominent forehead, delayed skeletal maturation, small genitalia.

Diabetes insipidus

Characterized by intense thirst and polyuria with inability to concentrate urine due to lack of ADH (central) or failure of response to circulating ADH (nephrogenic).

	Central DI	Nephrogenic DI
ETIOLOGY	Pituitary tumor, autoimmune, trauma, surgery, ischemic encephalopathy, idiopathic	Hereditary (ADH receptor mutation), 2° to hypercalcemia, hypokalemia, lithium, demeclocycline (ADH antagonist)
FINDINGS	↓ ADH Urine specific gravity < 1.006 Serum osmolality > 290 mOsm/kg Hyperosmotic volume contraction	Normal or ↑ ADH levels Urine specific gravity < 1.006 Serum osmolality > 290 mOsm/kg Hyperosmotic volume contraction
WATER DEPRIVATION TEST ^a	> 50% ↑ in urine osmolality only after administration of ADH analog	Minimal change in urine osmolality, even after administration of ADH analog
TREATMENT	Desmopressin acetate Hydration	HCTZ, indomethacin, amiloride Hydration, dietary salt restriction, avoidance of offending agent

^aNo water intake for 2–3 hr followed by hourly measurements of urine volume and osmolality and plasma Na⁺ concentration and osmolality. ADH analog (desmopressin acetate) is administered if serum osmolality > 295–300 mOsm/kg, plasma Na⁺ ≥ 145, or urine osmolality does not rise despite a rising plasma osmolality.

Syndrome of inappropriate antidiuretic hormone secretion

Characterized by:

- Excessive free water retention
- Euvolemic hyponatremia with continued urinary Na⁺ excretion
- Urine osmolality > serum osmolality

Body responds to water retention with ↓ aldosterone and ↑ ANP and BNP → ↑ urinary Na⁺ secretion → normalization of extracellular fluid volume → euvolemic hyponatremia. Very low serum Na⁺ levels can lead to cerebral edema, seizures. Correct slowly to prevent osmotic demyelination syndrome (formerly known as central pontine myelinolysis).

SIADH causes include:

- Ectopic ADH (eg, small cell lung cancer)
- CNS disorders/head trauma
- Pulmonary disease
- Drugs (eg, cyclophosphamide)

Treatment: fluid restriction, salt tablets, IV hypertonic saline, diuretics, conivaptan, tolvaptan, demeclocycline.

Increased urine osmolality during water deprivation test indicates psychogenic polydipsia.

Hypopituitarism

Undersecretion of pituitary hormones due to:

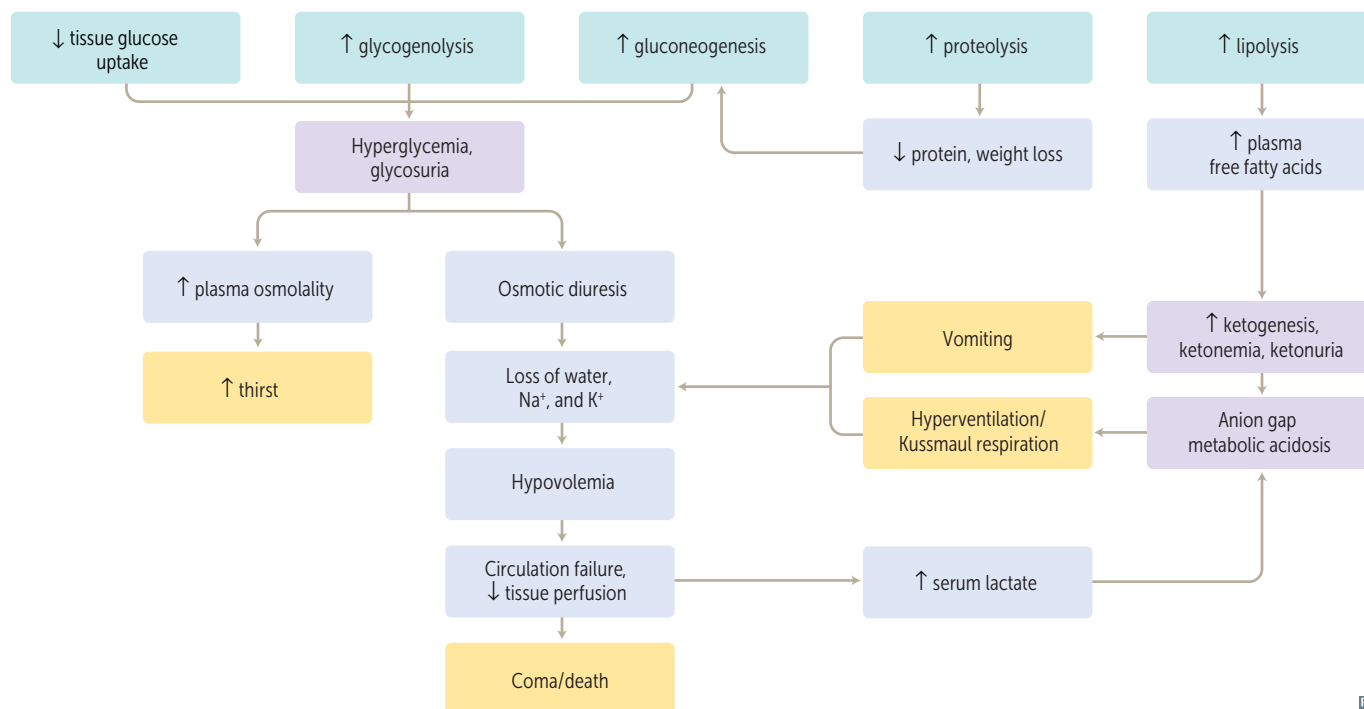
- Nonsecreting pituitary adenoma, craniopharyngioma
- **Sheehan syndrome**—ischemic infarct of pituitary following postpartum bleeding; pregnancy-induced pituitary growth → ↑ susceptibility to hypoperfusion. Usually presents with failure to lactate, absent menstruation, cold intolerance
- **Empty sella syndrome**—atrophy or compression of pituitary (which lies in the sella turcica), often idiopathic, common in obese women
- **Pituitary apoplexy**—sudden hemorrhage of pituitary gland, often in the presence of an existing pituitary adenoma. Usually presents with sudden onset severe headache, visual impairment (eg, bitemporal hemianopia, diplopia due to CN III palsy), and features of hypopituitarism.
- Brain injury
- Radiation

Treatment: hormone replacement therapy (corticosteroids, thyroxine, sex steroids, human growth hormone).

Diabetes mellitus

ACUTE MANIFESTATIONS	Polydipsia, polyuria, polyphagia, weight loss, DKA (type 1), hyperosmolar coma (type 2). Rarely, can be caused by unopposed secretion of GH and epinephrine. Also seen in patients on glucocorticoid therapy (steroid diabetes).		
CHRONIC COMPLICATIONS	Nonenzymatic glycation: - Small vessel disease (diffuse thickening of basement membrane) → retinopathy (hemorrhage, exudates, microaneurysms, vessel proliferation), glaucoma, neuropathy, nephropathy (nodular glomerulosclerosis, aka Kimmelstiel-Wilson nodules → progressive proteinuria [initially microalbuminuria; ACE inhibitors are renoprotective] and arteriolosclerosis → hypertension; both lead to chronic renal failure). - Large vessel atherosclerosis, CAD, peripheral vascular occlusive disease, gangrene → limb loss, cerebrovascular disease. MI most common cause of death. Osmotic damage (sorbitol accumulation in organs with aldose reductase and ↓ or absent sorbitol dehydrogenase): - Neuropathy (motor, sensory [glove and stocking distribution], and autonomic degeneration) - Cataracts		
DIAGNOSIS	TEST	DIAGNOSTIC CUTOFF	NOTES
	HbA _{1c}	≥ 6.5%	Reflects average blood glucose over prior 3 months
	Fasting plasma glucose	≥ 126 mg/dL	Fasting for > 8 hours
	2-hour oral glucose tolerance test	≥ 200 mg/dL	2 hours after consumption of 75 g of glucose in water

Insulin deficiency or severe insulin insensitivity



Type 1 vs type 2 diabetes mellitus

Variable	Type 1	Type 2
1° DEFECT	Autoimmune destruction of β cells (eg, due to glutamic acid decarboxylase antibodies)	$\uparrow$ resistance to insulin, progressive pancreatic β -cell failure
INSULIN NECESSARY IN TREATMENT	Always	Sometimes
AGE (EXCEPTIONS COMMONLY OCCUR)	< 30 yr	> 40 yr
ASSOCIATION WITH OBESITY	No	Yes
GENETIC PREDISPOSITION	Relatively weak (50% concordance in identical twins), polygenic	Relatively strong (90% concordance in identical twins), polygenic
ASSOCIATION WITH HLA SYSTEM	Yes (HLA-DR3 and -DR4)	No
GLUCOSE INTOLERANCE	Severe	Mild to moderate
INSULIN SENSITIVITY	High	Low
KETOACIDOSIS	Common	Rare
β -CELL NUMBERS IN THE ISLETS	$\downarrow$	Variable (with amyloid deposits)
SERUM INSULIN LEVEL	$\downarrow$	Variable
CLASSIC SYMPTOMS OF POLYURIA, POLYDIPSIA, POLYPHAGIA, WEIGHT LOSS	Common	Sometimes
HISTOLOGY	Islet leukocytic infiltrate	Islet amyloid polypeptide (IAPP) deposits

Diabetic ketoacidosis	One of the most feared complications of diabetes. Usually due to insulin noncompliance or $\uparrow$ insulin requirements from $\uparrow$ stress (eg, infection). Excess fat breakdown and $\uparrow$ ketogenesis from $\uparrow$ free fatty acids, which are then made into ketone bodies (β -hydroxybutyrate > acetoacetate). Usually occurs in type 1 diabetes, as endogenous insulin in type 2 diabetes usually prevents lipolysis.
SIGNS/SYMPTOMS	DKA is D eadly: D elirium/psychosis, K ussmaul respirations (rapid/deep breathing), A bdominal pain/nausea/vomiting, D ehydration. Fruity breath odor (due to exhaled acetone).
LABS	Hyperglycemia, $\uparrow$ H^+ , $\downarrow$ HCO_3^- ($\uparrow$ anion gap metabolic acidosis), $\uparrow$ blood ketone levels, leukocytosis. Hyperkalemia, but depleted intracellular K^+ due to transcellular shift from $\downarrow$ insulin and acidosis (therefore total body K^+ is depleted).
COMPLICATIONS	Life-threatening mucormycosis (usually caused by <i>Rhizopus</i> infection), cerebral edema, cardiac arrhythmias, heart failure.
TREATMENT	IV fluids, IV insulin, and K^+ (to replete intracellular stores); glucose if necessary to prevent hypoglycemia.

**Hyperosmolar
hyperglycemic state**

State of profound hyperglycemia-induced dehydration and ↑ serum osmolality, classically seen in elderly type 2 diabetics with limited ability to drink. Hyperglycemia → excessive osmotic diuresis → dehydration → eventual onset of HHNS. Symptoms: thirst, polyuria, lethargy, focal neurological deficits (eg, seizures), can progress to coma and death if left untreated. Labs: hyperglycemia (often > 600 mg/dL), ↑ serum osmolality (> 320 mOsm/kg), no acidosis (pH > 7.3, ketone production inhibited by presence of insulin). Treatment: aggressive IV fluids, insulin therapy.

Glucagonoma

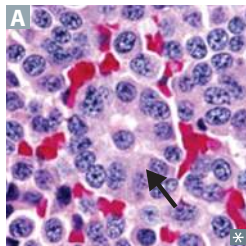
Tumor of pancreatic α cells → overproduction of glucagon. Presents with **d**ermatitis (necrolytic migratory erythema), **d**iabetes (hyperglycemia), **DVT**, **d**eclining weight, **d**epression. Treatment: octreotide, surgery.

Insulinoma

Tumor of pancreatic β cells → overproduction of insulin → hypoglycemia. May see Whipple triad: low blood glucose, symptoms of hypoglycemia (eg, lethargy, syncope, diplopia), and resolution of symptoms after normalization of glucose levels. Symptomatic patients have ↓ blood glucose and ↑ C-peptide levels (vs exogenous insulin use). ~ 10% of cases associated with MEN 1 syndrome. Treatment: surgical resection.

Somatostatinoma

Tumor of pancreatic δ cells → overproduction of somatostatin → ↓ secretion of secretin, cholecystokinin, glucagon, insulin, gastrin, gastric inhibitory peptide (GIP). May present with diabetes/glucose intolerance, steatorrhea, gallstones, achlorhydria. Treatment: surgical resection; somatostatin analogs (eg, octreotide) for symptom control.

Carcinoid syndrome

Rare syndrome caused by carcinoid tumors (neuroendocrine cells **A**; note prominent rosettes [arrow]), especially metastatic small bowel tumors, which secrete high levels of serotonin (5-HT). Not seen if tumor is limited to GI tract (5-HT undergoes first-pass metabolism in liver). Results in recurrent diarrhea, cutaneous flushing, asthmatic wheezing, right-sided valvular heart disease (tricuspid regurgitation, pulmonic stenosis). ↑ 5-hydroxyindoleacetic acid (5-HIAA) in urine, niacin deficiency (pellagra). Treatment: surgical resection, somatostatin analog (eg, octreotide).

Rule of 1/3s:

- 1/3** metastasize
- 1/3** present with 2nd malignancy
- 1/3** are multiple

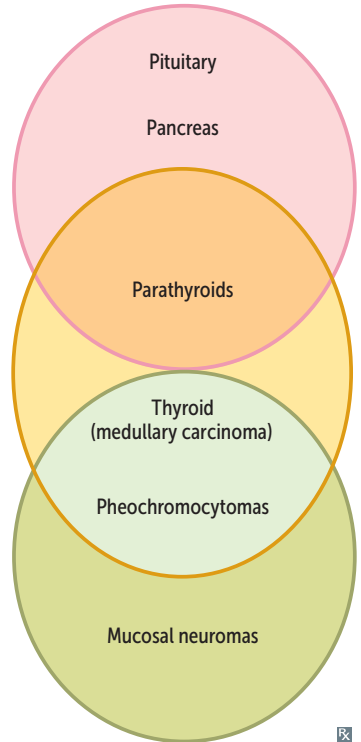
Most common malignancy in the small intestine.

**Zollinger-Ellison
syndrome**

Gastrin-secreting tumor (gastrinoma) of pancreas or duodenum. Acid hypersecretion causes recurrent ulcers in duodenum and jejunum. Presents with abdominal pain (peptic ulcer disease, distal ulcers), diarrhea (malabsorption). Positive secretin stimulation test: gastrin levels remain elevated after administration of secretin, which normally inhibits gastrin release. May be associated with MEN 1.

Multiple endocrine neoplasias

All **MEN** syndromes have autosomal **dominant** inheritance.
 “All **MEN** are **dominant**” (or so they think).

SUBTYPE	CHARACTERISTICS	COMMENTS
MEN 1	P ituitary tumors (prolactin or GH) P ancreatic endocrine tumors—Zollinger-Ellison syndrome, insulinomas, VIPomas, glucagonomas (rare) P arathyroid adenomas Associated with mutation of <i>MEN1</i> (menin, a tumor suppressor, chromosome 11)	
MEN 2A	P arathyroid hyperplasia Medullary thyroid carcinoma—neoplasm of parafollicular or C cells; secretes calcitonin; prophylactic thyroidectomy required P heochromocytoma (secretes catecholamines) Associated with mutation in <i>RET</i> (codes for receptor tyrosine kinase) in cells of neural crest origin	
MEN 2B	Medullary thyroid carcinoma P heochromocytoma Mucosal neuromas (oral/intestinal ganglioneuromatosis) Associated with marfanoid habitus; mutation in <i>RET</i> gene	

MEN 1 = 3 P's: Pituitary, **P**arathyroid, and **P**ancreas

MEN 2A = 2 P's: Parathyroids and **P**heochromocytoma

MEN 2B = 1 P: Pheochromocytoma

► ENDOCRINE—PHARMACOLOGY

Diabetes mellitus management

Treatment strategies:

- Type 1 DM—dietary modifications, insulin replacement
- Type 2 DM—dietary modifications and exercise for weight loss; oral agents, non-insulin injectables, insulin replacement
- Gestational DM (GDM)—dietary modifications, exercise, insulin replacement if lifestyle modification fails

DRUG CLASSES	CLINICAL USE	ACTION	RISKS/CONCERNS
Insulin preparations			
Insulin, rapid acting Lispro, aspart, glulisine	Type 1 DM, type 2 DM, GDM (postprandial glucose control).	Binds insulin receptor (tyrosine kinase activity). Liver: ↑ glucose stored as glycogen. Muscle: ↑ glycogen, protein synthesis; ↑ K ⁺ uptake. Fat: ↑ TG storage.	Hypoglycemia, lipodystrophy, rare hypersensitivity reactions.
Insulin, short acting Regular	Type 1 DM, type 2 DM, GDM, DKA (IV), hyperkalemia (+ glucose), stress hyperglycemia.		
Insulin, intermediate acting NPH	Type 1 DM, type 2 DM, GDM.		
Insulin, long acting Detemir, glargine	Type 1 DM, type 2 DM, GDM (basal glucose control).		
Oral drugs			
Biguanides Metformin	Oral. First-line therapy in type 2 DM, causes modest weight loss. Can be used in patients without islet function.	Inhibit hepatic gluconeogenesis and the action of glucagon. ↓ gluconeogenesis, ↑ glycolysis, ↑ peripheral glucose uptake (↑ insulin sensitivity).	GI upset; most serious adverse effect is lactic acidosis (thus contraindicated in renal insufficiency).
Sulfonylureas First generation: chlorpropamide, tolbutamide Second generation: glimepiride, glipizide, glyburide	Stimulate release of endogenous insulin in type 2 DM. Require some islet function, so useless in type 1 DM.	Close K ⁺ channel in β cell membrane → cell depolarizes → insulin release via ↑ Ca ²⁺ influx.	Risk of hypoglycemia ↑ in renal failure, weight gain. First generation: disulfiram-like effects. Second generation: hypoglycemia.
Glitazones/thiazolidinediones Pioglitazone, rosiglitazone	Used as monotherapy in type 2 DM or combined with above agents. Safe to use in renal impairment.	↑ insulin sensitivity in peripheral tissue. Binds to PPAR-γ nuclear transcription regulator. ^a	Weight gain, edema, HF, ↑ risk of fractures.

Diabetes mellitus management (continued)

DRUG CLASSES	CLINICAL USE	ACTION	RISKS/CONCERNS
Oral hypoglycemic drugs (continued)			
Meglitinides Nateglinide, repaglinide	Used as monotherapy in type 2 DM or combined with metformin.	Stimulate postprandial insulin release by binding to K ⁺ channels on β cell membranes (site differs from sulfonylureas).	Hypoglycemia ($\uparrow$ risk with renal failure), weight gain.
GLP-1 analogs Exenatide, liraglutide (sc injection)	Type 2 DM.	$\uparrow$ glucose-dependent insulin release, $\downarrow$ glucagon release, $\downarrow$ gastric emptying, $\uparrow$ satiety.	Nausea, vomiting, pancreatitis; modest weight loss.
DPP-4 inhibitors Linagliptin, saxagliptin, sitagliptin	Type 2 DM.	Inhibit DPP-4 enzyme that deactivates GLP-1, thereby $\uparrow$ glucose-dependent insulin release, $\downarrow$ glucagon release, $\downarrow$ gastric emptying, $\uparrow$ satiety.	Mild urinary or respiratory infections; weight neutral.
Amylin analogs Pramlintide (sc injection)	Type 1 DM, type 2 DM.	$\downarrow$ gastric emptying, $\downarrow$ glucagon.	Hypoglycemia (in setting of mistimed prandial insulin), nausea.
Sodium-glucose co-transporter 2 (SGLT2) inhibitors Canagliflozin, dapagliflozin, empagliflozin	Type 2 DM.	Block reabsorption of glucose in PCT.	Glucosuria, UTIs, vaginal yeast infections, hyperkalemia, dehydration (orthostatic hypotension), weight loss.
α-glucosidase inhibitors Acarbose, miglitol	Type 2 DM.	Inhibit intestinal brush-border α -glucosidases. Delayed carbohydrate hydrolysis and glucose absorption $\rightarrow$ $\downarrow$ postprandial hyperglycemia.	GI disturbances.

^aGenes activated by PPAR- γ regulate fatty acid storage and glucose metabolism. Activation of PPAR- γ $\uparrow$ insulin sensitivity and levels of adiponectin.

Thioamides

Propylthiouracil, methimazole.

MECHANISM	Block thyroid peroxidase, inhibiting the oxidation of iodide and the organification and coupling of iodine $\rightarrow$ inhibition of thyroid hormone synthesis. Propylthiouracil also blocks 5'-deiodinase $\rightarrow$ $\downarrow$ peripheral conversion of T ₄ to T ₃ .
CLINICAL USE	Hyperthyroidism. PTU blocks P eripheral conversion. PTU used in first trimester of pregnancy (due to methimazole teratogenicity); methimazole used in second and third trimesters of pregnancy (due to risk of PTU-induced hepatotoxicity).
ADVERSE EFFECTS	Skin rash, agranulocytosis (rare), aplastic anemia, hepatotoxicity. Methimazole is a possible teratogen (can cause aplasia cutis).

Levothyroxine, triiodothyronine

MECHANISM	Thyroid hormone replacement.
CLINICAL USE	Hypothyroidism, myxedema. Used off-label as weight loss supplements.
ADVERSE EFFECTS	Tachycardia, heat intolerance, tremors, arrhythmias.

Hypothalamic/pituitary drugs

DRUG	CLINICAL USE
ADH antagonists (conivaptan, tolvaptan)	SIADH, block action of ADH at V ₂ -receptor.
Desmopressin acetate	Central (not nephrogenic) DI, von Willebrand disease, sleep enuresis.
GH	GH deficiency, Turner syndrome.
Oxytocin	Stimulates labor, uterine contractions, milk let-down; controls uterine hemorrhage.
Somatostatin (octreotide)	Acromegaly, carcinoid syndrome, gastrinoma, glucagonoma, esophageal varices.

Demeclocycline

MECHANISM	ADH antagonist (member of tetracycline family).
CLINICAL USE	SIADH.
ADVERSE EFFECTS	Nephrogenic DI, photosensitivity, abnormalities of bone and teeth.

Fludrocortisone

MECHANISM	Synthetic analog of aldosterone with little glucocorticoid effects.
CLINICAL USE	Mineralocorticoid replacement in 1° adrenal insufficiency.
ADVERSE EFFECTS	Similar to glucocorticoids; also edema, exacerbation of heart failure, hyperpigmentation.

Cinacalcet

MECHANISM	Sensitizes Ca ²⁺ -sensing receptor (CaSR) in parathyroid gland to circulating Ca ²⁺ → ↓ PTH.
CLINICAL USE	1° or 2° hyperparathyroidism.
ADVERSE EFFECTS	Hypocalcemia.